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IEEE International Symposium on Biomedical Imaging 2024 : Submission (1187) has been created.

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Hello,

The following submission has been created.

Track Name: IEEE ISBI 2024

Paper ID: 1187

Paper Title: RIB FRACTURE DETECTION USING DEEP LEARNING ON CHEST X-RAY IMAGES

Abstract:
Rib fractures represent the most common injury resulting from thoracic trauma. Such acute fractures are often missed on radiographs. We explore the use of convolutional neural network models to classify individual chest X-ray images as either containing a rib fracture or containing no fracture. Such a model can be used as a tool to aid clinicians in confirming a rib fracture diagnosis using chest X-ray images captured after a trauma and also to flag radiographs that feature incidental rib fractures. The EfficientNet B2 model achieved a validation accuracy of 82% and a validation AUC of 86%. These results show deep learning models can confirm rib fractures within chest radiographs to a strong degree of accuracy. This suggests it is feasible to develop a tool to help clinicians detect these fractures, saving the health system on downstream costs associated with missed fractures and increasing efficiency in the clinic.

Created on: Fri, 24 Nov 2023 21:12:30 GMT

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Primary Subject Area: Computational and statistical image processing and analysis

Secondary Subject Areas: Classification
Computational Imaging
Deep learning
Imaging modalities - X-ray imaging
Machine learning

Submission Files: RIB FRACTURE DETECTION USING DEEP LEARNING ON CHEST X-RAY IMAGES_11.24.23_vF.pdf (192 Kb, Fri, 24 Nov 2023 21:11:31 GMT)

Submission Questions Response:
1. Statement of Novelty/Impact
This research shows that deep learning models can be trained to detect the presence of a rib fracture in a chest X-ray with strong accuracy. Our approach is unique in that the entire image is processed at once by the neural network rather than focusing on an image subsection.
2. Review Conflicts
Agreement accepted

3. Declaration of Other Conflicts
N/A
4. Student Survey
Not a student
5. ISBI Best Paper Award
Don't know
6. Agreement
Agreement accepted
7. Information Transfer
Yes
8. Human and/or Animal Subjects
No
9. Compliance with Ethical Standards
Agreement accepted
10. Conflicts of Interest Disclosure
Agreement accepted
11. Form of Attendence
Yes

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